

15

MEASUREMENT AND SPACE

CIRCLE GEOMETRY

OPTIONAL PATH TOPIC
FOR HSC MATHEMATICS EXTENSION

The ancient Greek philosopher, Thales of Miletus (624–546 BCE), is believed to have been the first mathematician and scientist to attempt to explain things by reasoning and deduction rather than by resorting to mythology and superstition. He lived before Pythagoras and proved the first theorem in mathematics: that any angle drawn on the circumference of a semicircle is a right angle. This rule is often called **Thales' theorem.**



Chapter outline

	Working mathematically				
15.01 Parts of a circle*	U	F			C
15.02 Chord properties of circles*	U	F		R	C
15.03 Angle properties of circles*	U	F		R	C
15.04 Tangent and secant properties of circles*	U	F		R	C
15.05 Proofs using circle theorems*	U	F		R	C

***PATH**

Wordbank

collinear points Points that lie on a straight line.

converse A rule or statement turned back-to-front; the reverse statement

cyclic quadrilateral A quadrilateral bounded by a circle, whose vertices lie on the circle's circumference

intercept The distance between the points where a line intersects other lines or curves

perpendicular bisector An interval or line that bisects another interval at right angles

secant A line that intersects a curve at 2 points

subtend an angle To sit opposite an angle and 'hold' its arms

tangent A line that touches a curve at one point but does not cross it

Quiz
Wordbank 15

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

Video (1):
15.03, 15.04 Circle geometry

Quiz (1):
- Wordbank 15

Worksheets (6):
15.01 Parts of a circle
15.02 A page of circles
15.03 A page of circles • Angle problems with algebra
15.04 Circle geometry cards • Circle geometry review • A page of circles

Puzzle (1):
15.03 Finding angles in circles

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

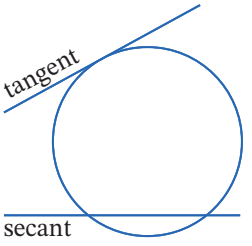
In this chapter you will:

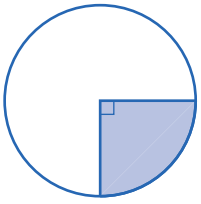
- (PATH) identify the parts of a circle
- (PATH) prove chord and angle properties of circles and use them to find unknown lengths and angles in circles
- (PATH) prove tangent and secant properties of circles and use them to find unknown lengths and angles in circles

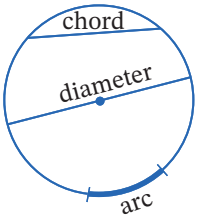
15.01 Parts of a circle

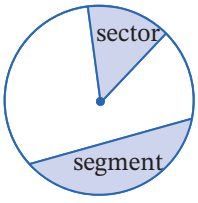
PATH

Worksheet
Parts of a circle




Quadrant





A **tangent** is a line touching the circle at one point.

A **secant** is a line that cuts the circle at 2 points.

A **quadrant** is quarter of a circle.

A **chord** is an interval joining 2 points on the circle.

A **diameter** is a chord that passes through the centre.

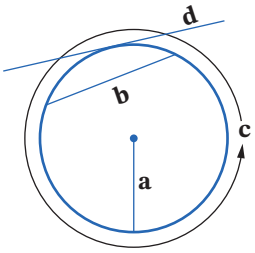
An **arc** is a part of the circumference of the circle.

A **sector** is part of a circle bounded by 2 radii and an arc.

A **segment** is part of a circle cut off by a chord.

EXERCISE 15.01 ANSWERS ON P. 687

Parts of a circle U F C

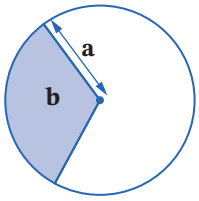


1 Name each part of the circle marked by a letter. C

2 **a** Draw a sector and a segment in a circle.

b What is the difference between a sector and a segment? C

3 State the mathematical rule connecting d (diameter) and r (radius) as an algebraic formula. C

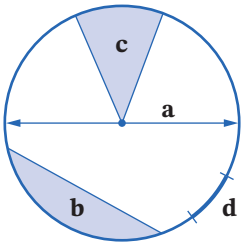


4 Which parts of a circle are labelled **a** and **b**? Select the correct answer **A**, **B**, **C** or **D**. C

A diameter and segment **B** radius and segment

C diameter and sector **D** radius and sector

5 Name each part of the circle marked by a letter. C



6 Write the correct word to match each description. C

a The distance from the centre of a circle to its side

b Quarter of a circle

c A line that touches the outside of a circle once

d An interval from one side of a circle to the other side, through the circle's centre

e An interval from one side of the circle to the other side, not through the centre

f Part of the circumference of a circle

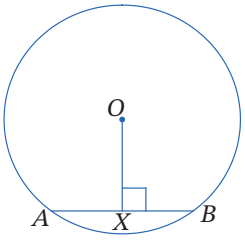
g The area inside a circle formed by 2 radii and an arc

h The complete distance around a circle

i The area inside a circle formed by a chord and an arc

Technology

Perpendicular to a chord



1 Use dynamic geometry software to draw a circle with a centre, O.

2 Mark 2 points A and B on the circumference of the circle.

3 Join the 2 points to make a chord AB.

4 Select the centre of the circle and the chord.

5 Draw a **perpendicular line** from the centre to the chord, and make the point of intersection X.

6 Measure AX and BX. What do you notice?

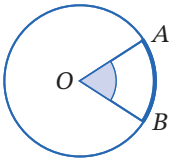
7 Move points B and C to draw different chords. Measure AX and BX again. What do you notice again?

15.02 Chord properties of circles

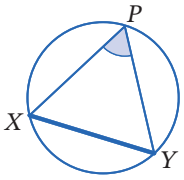
PATH

Worksheet
A page of circles

When an arc or chord of a circle **subtends** an angle, it sits opposite or under the angle and 'holds' the angle's arms. The angle stands on the ends of the arc or chord. The angle is **subtended by** the arc or chord.



The arc AB subtends the angle at O.



The chord XY subtends the angle at P.

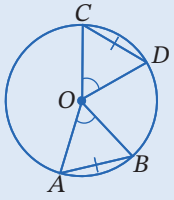
A **theorem** is a rule or statement that can be proved from basic principles. We will now look at circle theorems relating to chords that can be proved by congruent triangles. These theorems can also be demonstrated using dynamic geometry software or pencil-and-paper constructions.

Equal chords subtend equal angles

Theorem 1: Chords of equal length in a circle subtend equal angles at the centre of the circle.

If $AB = CD$,

$\angle AOB = \angle COD$.



Proof:

In $\triangle AOB$ and $\triangle COD$:

$AB = CD$	(given)
$OA = OC$	(equal radii)
$OB = OD$	(equal radii)
$\therefore \triangle AOB \equiv \triangle COD$	(SSS)
$\therefore \angle AOB = \angle COD$	(matching angles of congruent triangles)

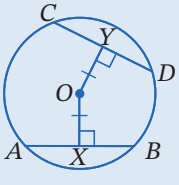
So chords of equal length subtend equal angles at the centre of a circle.

Equal chords equidistant from centre

Theorem 2: Chords of equal length in a circle are equidistant from the centre of the circle.

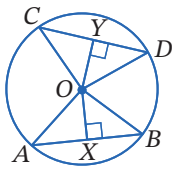
If $AB = CD$,

then $OX = OY$.



Proof:

Draw radii OA, OB, OC and OD.



In $\triangle AOX$ and $\triangle COY$:

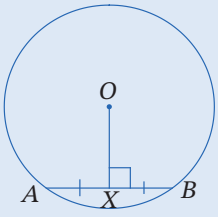
$OA = OC$	(equal radii)
$\angle A = \angle C$	(matching angles of congruent $\triangle AOB$ and $\triangle COD$ proved in Theorem 1)
$\angle OXA = \angle OYC = 90°$	(given)
$\therefore\ \triangle AOX \equiv \triangle COY$	(AAS)
$\therefore\ OX = OY$	(matching sides of congruent triangles)

So equal chords are equidistant from the centre of a circle.

Perpendicular from centre to chord

Theorem 3: The perpendicular from the centre of a circle to a chord bisects the chord.

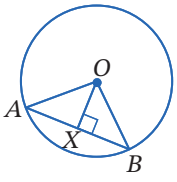
The **converse** of this is also true: The line from the centre of a circle to the midpoint of a chord is perpendicular to the chord.



The word 'converse' means to 'turn around' or 'reverse'. So the converse of a theorem is the theorem written 'back-to-front'.

Proof:

Draw radii OA and OB.



In $\triangle AOX$ and $\triangle BOX$:

$OA = OB$ (equal radii)

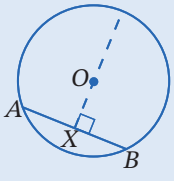
OX is common.

$\angle OXA = \angle OXB = 90°$	($OX \perp AB$)
$\therefore\ \triangle AOX \equiv \triangle BOX$	(RHS)
$\therefore\ AX = BX$	(matching sides of congruent triangles)

So the perpendicular from the centre of a circle to a chord bisects the chord.

Perpendicular bisector of a chord

Theorem 4: The perpendicular bisector of a chord in a circle passes through the centre of the circle.



Proof:

The perpendicular from O to AB bisects AB. (proved in Theorem 3)

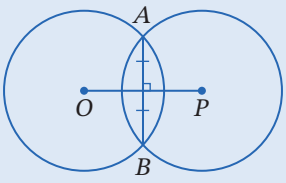
$\therefore$ OX is the perpendicular bisector of AB.

$\therefore$ The perpendicular bisector of AB passes through the centre O.

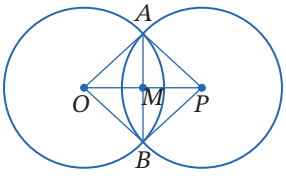
PATH

Common chords

Theorem 5: When 2 circles intersect, the line joining their centres bisects their common chord at right angles.



Proof:



Draw radii OA, OB, PA and PB and let OP and AB bisect at M.

In $\triangle OAP$ and $\triangle OBP$:

$OA = OB$ (equal radii)

$PA = PB$ (equal radii)

OP is common.

$\therefore \triangle OAP \equiv \triangle OBP$ (SSS)

$\therefore \angle AOM = \angle BOM$ (matching angles of congruent triangles)

In $\triangle OAM$ and $\triangle OBM$:

$OA = OB$ (equal radii)

OM is common.

$\therefore \angle AOM = \angle BOM$ (proved above)

$\therefore \triangle OAM \equiv \triangle OBM$ (SAS)

$\therefore AM = MB$ (matching sides of congruent triangles)

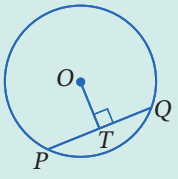
$\therefore AB \perp OP$ (line from the centre of a circle to the midpoint of a chord)

So the perpendicular from the centre of a circle to a chord bisects the chord.

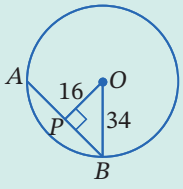
Example 1

In each diagram, O is the centre of a circle.

a If $PQ = 35$ cm, find PT.



b If $OP = 16$ cm, find AB.



Solution

a $PT = \frac{1}{2}(PQ)$ (the perpendicular from the centre bisects the chord)

$= \frac{1}{2} \times 35$

$= 17.5$ cm

 9780170479660

b $PB^2 = 34^2 - 16^2$

$= 900$

$PB = \sqrt{900}$

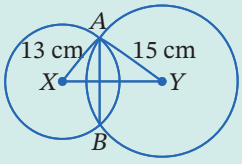
$= 30$ cm

$AB = 2 \times PB$ (the perpendicular from the centre bisects the chord)

$= 60$ cm

Example 2

The length AB of the common chord of 2 circles is 24 cm. If the radii of the circles are 13 cm and 15 cm, find the distance between their centres.



$AB = 24$ cm

Solution

We need to find XY.

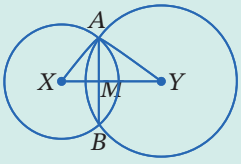
Let XY and AB cross at M.

$XY \perp AB$ (common chord of circles)

$AM = \frac{1}{2} \times AB$

$= \frac{1}{2} \times 24$

$= 12$ cm



$XM^2 = 13^2 - 12^2 = 25$ by Pythagoras' theorem

$XM = \sqrt{25} = 5$ cm

$YM^2 = 15^2 - 12^2 = 81$

$YM = \sqrt{81} = 9$ cm

$\therefore XY = 5 + 9 = 14$ cm

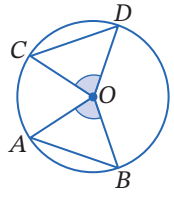
The distance between the centres is 14 cm.

EXERCISE 15.02 ANSWERS ON P. 688

Chord properties of circles U F R C

EXAMPLE 1

1 The converse of Theorem 1 is: 'Chords subtending equal angles at the centre of a circle have equal lengths'. R C

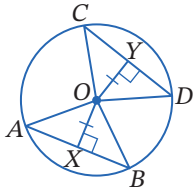


$\angle AOB = \angle COD$

a Prove that $\triangle AOB \equiv \triangle COD$, given that $\angle AOB = \angle COD$.

b Hence, prove that $AB = CD$.

2 The converse of Theorem 2 is: 'Chords that are equidistant from the centre of a circle are equal'. R C



$OX = OY$

a Prove that $\triangle OXA \equiv \triangle OYC$ and therefore $AX = CY$.

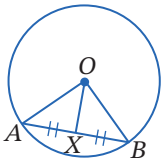
b Prove that $\triangle OXB \equiv \triangle OYD$ and therefore $BX = DY$.

c Hence show that $AB = CD$.

Foundation Standard Complex

PATH

3 The converse of Theorem 3 is: 'The line from the centre of a circle to the midpoint of a chord is perpendicular to the chord.' R C



$AX = BX$

a Prove that $\triangle OXA \equiv \triangle OXB$ and therefore $\angle OXA = \angle OXB$.

b Prove that $OX \perp AB$.

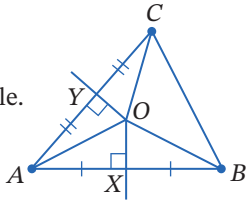
4 **a** Draw a circle and construct any 2 non-parallel chords. R C

b Bisect these chords.

c Through what point must each perpendicular bisector pass? (Use Theorem 4.)

d Explain why the point of intersection between the 2 perpendicular bisectors is the centre of the circle.

5 The diagram shows the perpendicular bisectors of 2 sides AB and CA of a triangle, intersecting at O. It can be proved that O is the centre of a circle passing through all 3 vertices of the triangle.



The full statement on this theorem is: 'Given any 3 non-collinear* points, the point of intersection of the perpendicular bisectors of any 2 sides of the triangle formed by the 3 points is the centre of the circle through all 3 points.'

(*Non-collinear means *not* on the same straight line.)

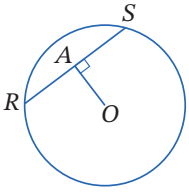
Prove this theorem as follows: R C

a Prove that $\triangle AOY \equiv \triangle COY$ and therefore $OA = OC$.

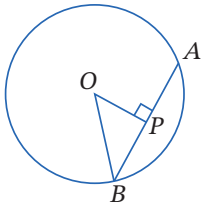
b Prove that $\triangle AOX \equiv \triangle BOX$ and therefore $OA = OB$.

c Hence prove that the circle with centre O and radius OA also passes through B and C.

6 This circle has a radius of 7 cm and the chord RS is 8 cm. Find, correct to 2 decimal places, the length of OA. R



7 In the diagram, $AB \perp OP$. If $OB = 32.5$ cm and $OP = 12.5$ cm, find AB. R

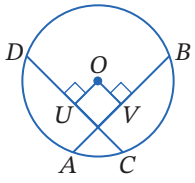


8 In each problem, state which chord theorem you use. R C

a $OU = OV$

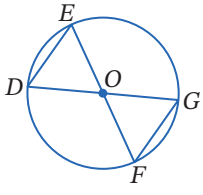
$AB = 9$ m

Find the length of UC.



b $GF = 12$ m

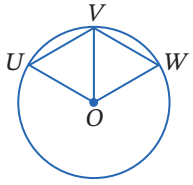
Find the length of DE.



c $UV = VW$

$\angle VOW = 64°$

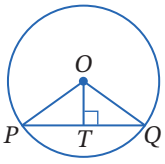
Find the size of $\angle UVO$.



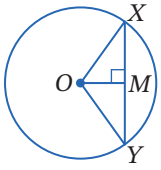
Foundation Standard Complex

9780170479660

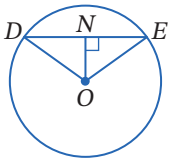
d $OT \perp PQ$
$OQ = 17$ mm
$OT = 8$ mm
Find PQ.



e $OM \perp XY$
$XY = 40$ cm
$OX = 29$ cm
Find OM.



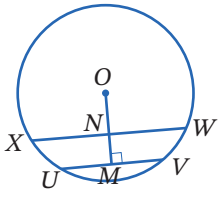
f $DE \perp ON$
$ON = 18$ m
$DE = 36$ m
Find the radius.



9 Find the radius of a circle in which a chord of length 96 cm is 20 cm from the centre. R

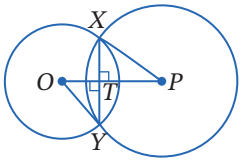
10 On a clear day, you can see 11.5 km in any direction from a particular lookout. If a straight road is 6.9 km from the lookout, what length of the road can be seen from the lookout? R

11 In the diagram, $OM \perp UV$, $XW \parallel UV$, $UV = 24$ cm and $XW = 32$ cm. If the radius of the circle is 20 cm, find the length of MN. R

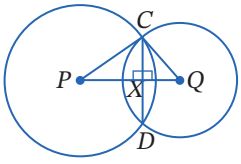


EXAMPLE 2

12 O and P are the centres of circles. If $OY = 40$ cm, $XP = 51$ cm and $XY = 48$ cm, find the distance between the centres. R



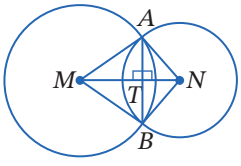
13 P and Q are the centres of circles. If $DC = 32$ cm, $PX = 30$ cm and $XQ = 12$ cm, find the radius of each circle. R



14 M and N are the centres of circles. $MT = 36$ cm, $NT = 20$ cm and $AM = 39$ cm. Find: R

a AB

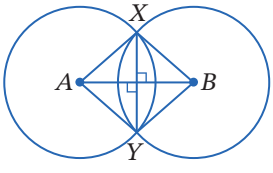
b the area of kite $ANBM$



15 A and B are the centres of circles with the same radius. $XY = 40$ cm and $AB = 96$ cm. Find: R C

a the radius of each circle

b the area of rhombus $AXBY$



Foundation | Standard | Complex

15.03 Angle properties of circles

PATH

Worksheets
A page of circles

Angle problems with algebra

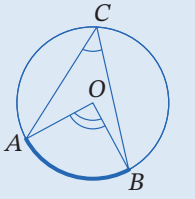
Puzzle
Finding angles in circles

We will now look at proofs of theorems relating to angles in a circle. These theorems can also be demonstrated using dynamic geometry software or pencil-and-paper constructions.

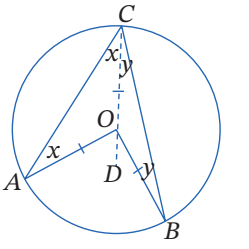
Angles at circumference and centre

Theorem 6: The angle at the centre of a circle is twice the angle at the circumference, standing on the same arc.

$$\angle AOB = 2 \times \angle ACB$$



$\angle AOB$ is the **angle at the centre** and $\angle ACB$ is the **angle at the circumference.** Both angles stand on the same arc AB.



Proof:

Draw radius CO and produce (extend) to point D.

Let $\angle ACD = x$ and $\angle BCD = y$.

$\therefore \angle ACB = x + y$

$OA = OC = OB$ (equal radii)

$\therefore \angle OAC = x$ (equal angles in isosceles $\triangle OAC$)

Similarly, $\angle OBC = y$ (equal angles in isosceles $\triangle OBC$)

$\angle AOD = x + x$ (exterior angle of $\triangle OAC$)

$= 2x$

Similarly, $\angle BOD = y + y$ (exterior angle of $\triangle OBC$)

$= 2y$

$\therefore \angle AOB = 2x + 2y$

$= 2(x + y)$

$= 2 \times \angle ACB$

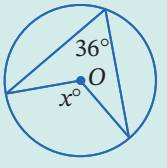
So the angle at the centre of the circle is twice the angle at the circumference, standing on the same arc.

Video
Circle geometry

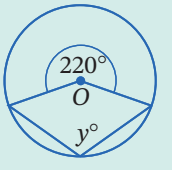
Example 3

Find the value of each variable.

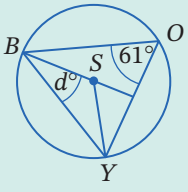
a



b



c



9780170479660

Solution

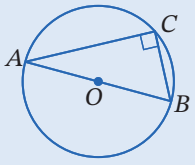
a $x = 2 \times 36$ (angle at centre is twice angle at circumference)
$= 72$

b $y = \frac{1}{2} \times 220$ (angle at centre is twice angle at circumference)
$= 110$

c $\angle BSY = 2 \times 61°$ (angle at centre is twice angle at circumference)
$= 122°$
$SB = SY$ (equal radii)
$\therefore \angle SBY = \angle SYB = d$ (equal angles in isosceles $\triangle SYB$)
$2d + 122 = 180$ (angle sum of $\triangle SYB$)
$2d = 58$
$d = 29$

Angle in a semicircle (Thales' theorem)

Theorem 7: The angle in a semicircle is a right angle.



$\angle ACB = 90°$

Proof:

$\angle AOB = 180°$ (a straight angle)

$\angle ACB = \frac{1}{2}$ of $\angle AOB$ (angle at centre is twice angle at circumference)

$= \frac{1}{2} \times 180°$

$= 90°$

The angle in a semicircle is a right angle.

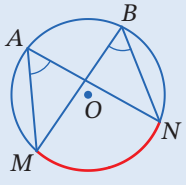
Angles in the same segment

Theorem 8: Angles at the circumference of a circle, standing on the same arc, are equal.

Or put another way, angles in the same segment are equal.

$\angle A = \angle B$

Angles $\angle A$ and $\angle B$ both stand on the same arc MN.



PATH

Proof:

Join radii OM and ON.

Let $\angle MON = 2x$

$\therefore \angle A = \frac{1}{2} \times \angle MON$ (angle at the centre is twice the angle at the circumference)

$= \frac{1}{2} \times 2x$

$= x$

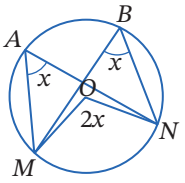
Similarly, $\angle B = \frac{1}{2} \times \angle MON$

$= \frac{1}{2} \times 2x$

$= x$

$\therefore \angle A = \angle B$

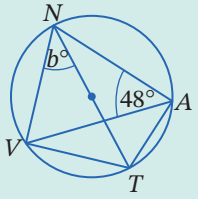
So angles standing on the same arc are equal.



Video
Circle geometry

Example 4

Find b, giving reasons.



Solution

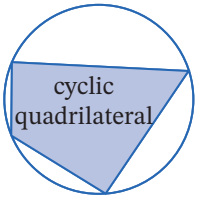
$\angle NAT = 90°$ (angle in a semicircle)

$\therefore \angle VAT = 90° - 48°$

$= 42°$

$\therefore b = 42$ (angles standing on the same arc)

A **cyclic quadrilateral** is a quadrilateral whose 4 vertices lie on the circumference of a circle.

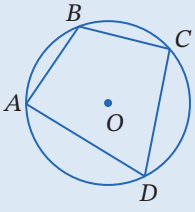


Cyclic quadrilaterals

Theorem 9: The opposite angles in a cyclic quadrilateral are supplementary.

$\angle A + \angle C = 180°$

$\angle B + \angle D = 180°$



Proof:

Draw radii OA and OC.

Let $\angle B = x°$ and $\angle D = y°$.

$\therefore$ Reflex $\angle AOC = 2x°$ (angle at centre is twice angle at the circumference)

$\therefore$ Obtuse $\angle AOC = 2y°$ (angle at centre is twice angle at the circumference)

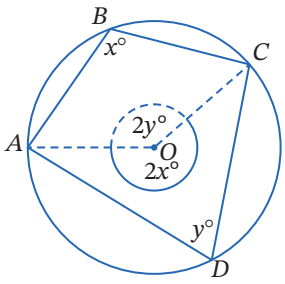
$2x + 2y = 360$ (angles at a point)

$x + y = 180$

$\therefore \angle B + \angle D = 180°$

Hence, $\angle A + \angle C = 180°$ (angle sum of a quadrilateral)

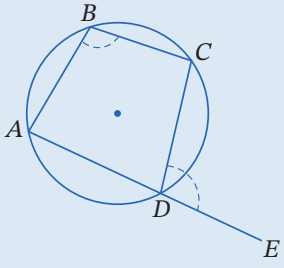
So the opposite angles of a cyclic quadrilateral are supplementary.



Exterior angle of a cyclic quadrilateral

Theorem 10: The exterior angle at a vertex of a cyclic quadrilateral is equal to the interior opposite angle.

$$\angle CDE = \angle B$$



Proof:

Let $\angle B = x°$.

$\angle ADC = 180° - x°$ (opposite angles of a cyclic quadrilateral)

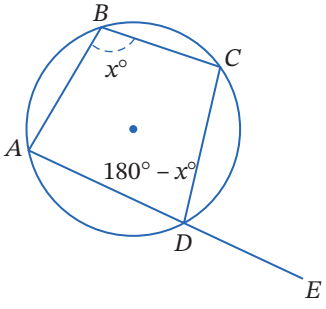
$\angle CDE = 180° - (180° - x°)$ (angles on a straight line)

$= 180 - 180 + x°$

$= x°$

$\therefore \angle ABC = \angle CDE = x°$

The exterior angle of a cyclic quadrilateral is equal to the interior opposite angle.



Example 5

Find p and q, giving reasons.

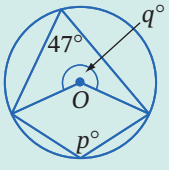
Solution

$p = 180 - 47$ (opposite angles of a cyclic quadrilateral)

$= 133$

$q = 2 \times 133$ (angle at centre is twice angle at circumference)

$= 266$



PATH

Video
Circle geometry

Example 6

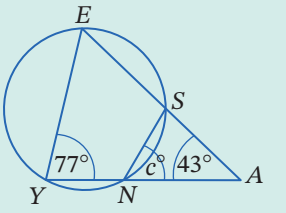
Find c, giving reasons.

Solution

$\angle ASN = 77°$ (exterior angle of cyclic quadrilateral $YESN$)

$c = 180 - 43 - 77$ (angle sum of $\triangle SAN$)

$= 60$



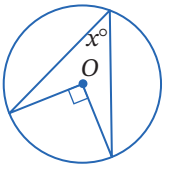
EXERCISE 15.03 ANSWERS ON P. 688

Angle properties of circles U F R C

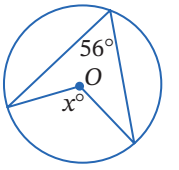
EXAMPLE 3

1 Find x, giving reasons, if O is the centre of the circle. R C

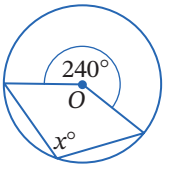
a



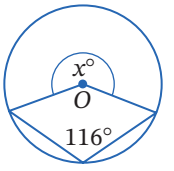
b



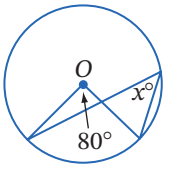
c



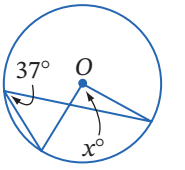
d



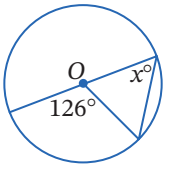
e



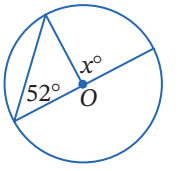
f



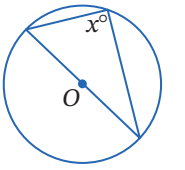
g



h



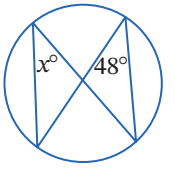
i



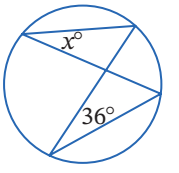
EXAMPLE 4

2 Find x, giving reasons. R C

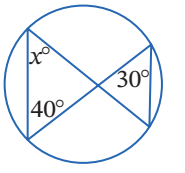
a



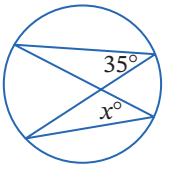
b



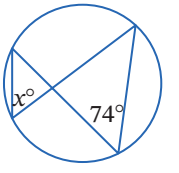
c



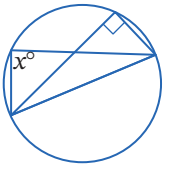
d



e



f



Foundation Standard Complex

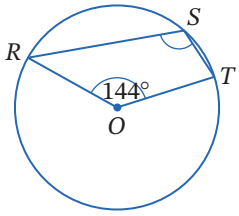
9780170479660

PATH

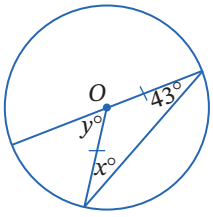
15.03

3 Find the value of each variable, giving reasons. R C

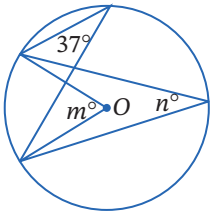
a



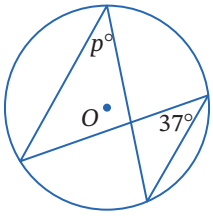
b



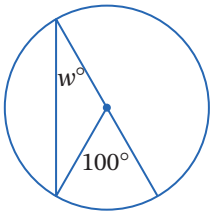
c



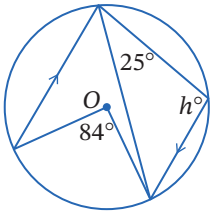
d



e



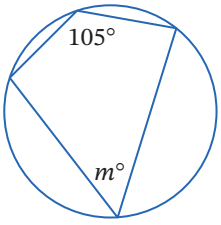
f



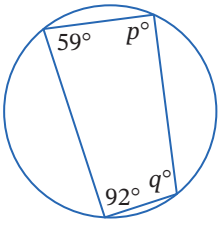
4 Find the value of each variable, giving reasons. R C

EXAMPLE 5

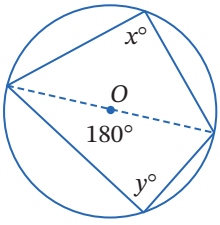
a



b



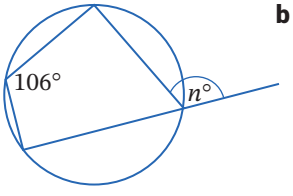
c



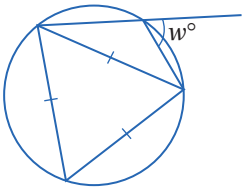
5 Find the value of each variable, giving reasons. R C

EXAMPLE 6

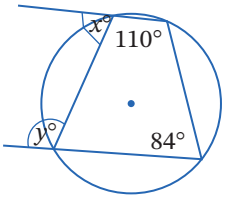
a



b

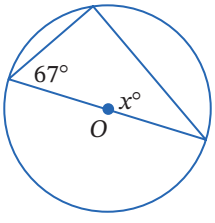


c

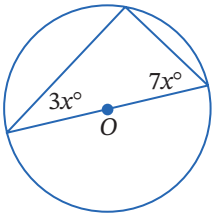


6 Find x, giving reasons. R C

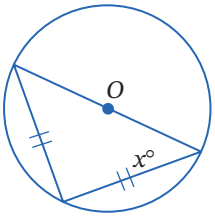
a



b



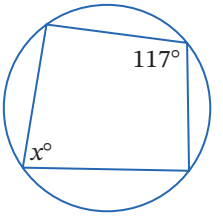
c



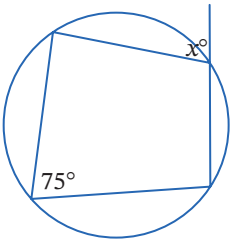
Foundation | Standard | Complex

PATH

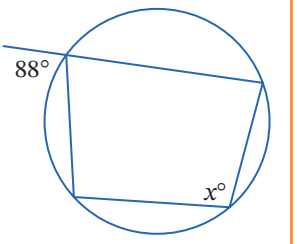
d



e

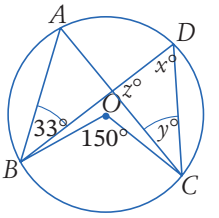


f

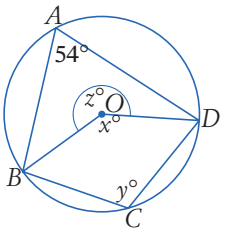


7 Find the value of each variable, giving reasons. R C

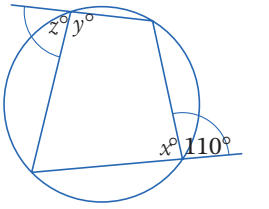
a



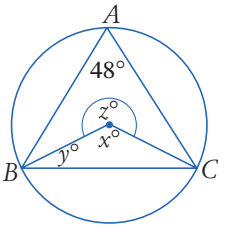
b



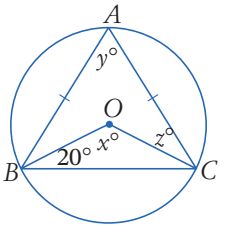
c



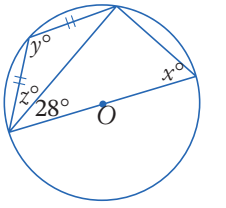
d



e

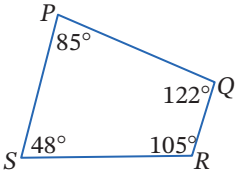


f

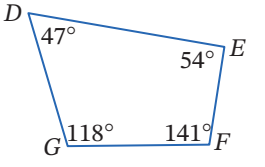


8 Which of these is a cyclic quadrilateral? Why? R C

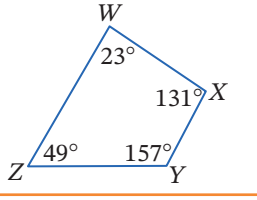
A



B



C



15.04 Tangent and secant properties of circles

PATH

Worksheet
Circle geometry cards

A **tangent** to a circle is a line that touches the circle at only one point.
The point of intersection is called the **point of contact**.

At any point on a circle, there is only one tangent to the circle at that point.

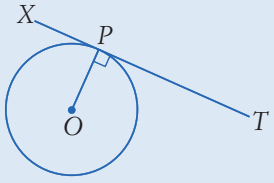
Foundation Standard Complex

9780170479660

Tangent theorems

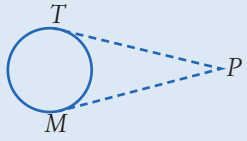
Theorem 11: A tangent to a circle is perpendicular to the radius drawn to the point of contact.

$\angle OPT = 90°$

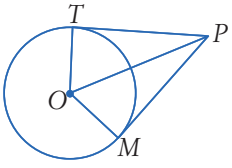


Theorem 12: The 2 tangents drawn to a circle from an external point are equal in length.

$PT = PM.$



Proof of Theorem 12:



Draw radii OT and OM, and join OP.

In $\triangle PTO$ and $\triangle PMO$:

OP is common.

$OT = OM$ (equal radii)

$\angle OTP = \angle OMP = 90°$ (angle between tangent and radius)

$\therefore \triangle PTO \equiv \triangle PMO$ (RHS)

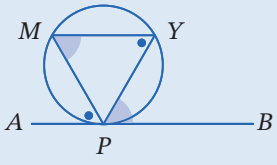
$\therefore PT = PM$ (matching sides of congruent triangles)

So tangents drawn from an external point are equal in length.

Alternate segment theorem

Theorem 13: The angle between a tangent and a chord drawn to the point of contact is equal to the angle in the **alternate segment.**

$\angle YPB = \angle M$



(To be proved in question **1** of the next exercise.)

AB is a tangent to the circle. The chord PY divides the circle into 2 segments.

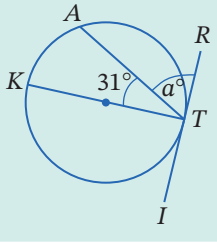
$\angle M$ is in the alternate (opposite) segment to $\angle YPB$.

Similarly, $\angle Y$ is in the alternate segment to $\angle APM$, so $\angle APM = \angle Y$.

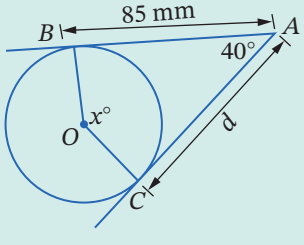
Example 7

Find the value of each variable, giving reasons.

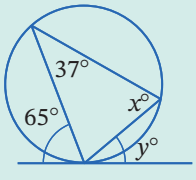
a



b



c



PATH

Worksheets
Circle geometry review

A page of circles

15.04

Video
Circle geometry

PATH

Solution

a $\angle KTR = 90°$ (angle between a tangent and a radius)

$a + 31 = 90$

$a = 59$

b $d = 85$ (tangents from an external point)

$\angle ABO = \angle ACO = 90°$ (angle between a tangent and a radius)

$\therefore x + 90 + 40 + 90 = 360$ (angle sum of a quadrilateral)

$x + 220 = 360$

$x = 140$

c $x = 65$ (angle in alternate segment)

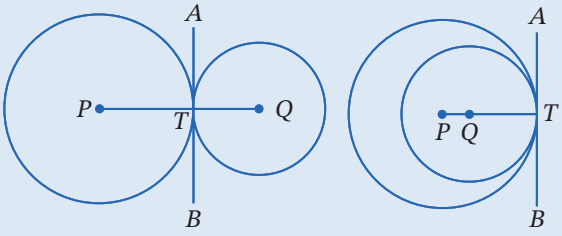
$y = 37$ (angle in alternate segment)

Touching circles

Theorem 14: When 2 circles touch, their centres and the point of contact are **collinear**.

The circles have a common tangent AB.

P, T and Q lie on a **straight** line



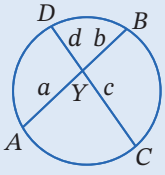
(To be proved in question **2** of the next exercise.)

When a line crosses another line or curve at 2 points, the distance between the points is called an **intercept**.

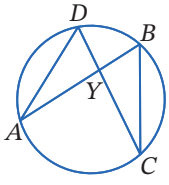
Intercepts of intersecting chords

Theorem 15: The products of the intercepts of 2 intersecting chords are equal.

$$ab = cd$$



a, b, c and d are called **intercepts**.



Proof:

Draw AD and CB.

In $\triangle ADY$ and $\triangle CBY$:

$\angle D = \angle B$ (angles in the same segment)

$\angle A = \angle C$ (angles in the same segment)

$\therefore \triangle ADY \,|||\, \triangle CBY$ (equiangular)

9780170479660

$\therefore \frac{AY}{CY} = \frac{YD}{YB}$ (matching sides of similar triangles)

$\therefore AY \times YB = CY \times YD$

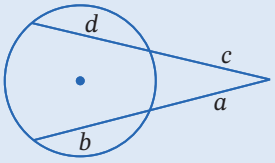
$\therefore ab = cd$

Therefore, the products of intercepts of 2 intersecting chords are equal.

Intersecting secants and the tangent-secant theorem

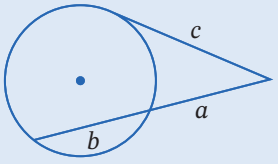
Theorem 16: The product of the intercepts of 2 intersecting secants to a circle from an external point are equal.

$$a(a + b) = c(c + d)$$



Theorem 17: The square of a tangent to a circle from an external point equals the product of the intercepts of any secant from the point.

$$c^2 = a(a + b)$$

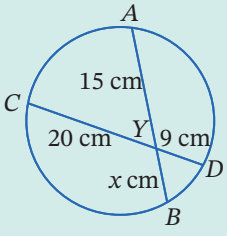


(To be proved in question **3** of the next exercise.)

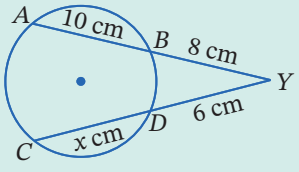
Example 8

Find x, giving reasons.

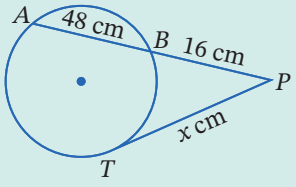
a



b



c



Solution

a $x \times 15 = 9 \times 20$ (intercepts of intersecting chords)

$15x = 180$

$x = 12$

b $6 \times (6 + x) = 8 \times 18$ (intercepts of intersecting secants)

$36 + 6x = 144$

$6x = 108$

$x = 18$

c $x^2 = 16 \times (16 + 48)$ (tangent-secant theorem)

$= 1024$

$x = \sqrt{1024}$

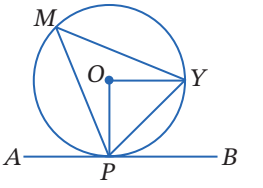
$= 32$

PATH

EXERCISE 15.04 ANSWERS ON P. 688

Tangent and secant properties of circles U F R C

1 AB is a tangent to a circle at P, with centre O. R C



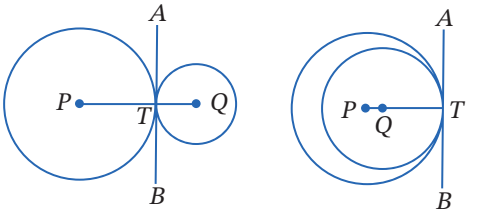
a If $\angle YPB$ is an angle between a tangent and chord, what is the angle in the alternate segment?

b If $\angle YPB = x°$, why is $\angle OPY = 90° - x°$?

c Show that $\angle POY = 2x°$.

d Hence show that $\angle PMY = \angle YPB$.

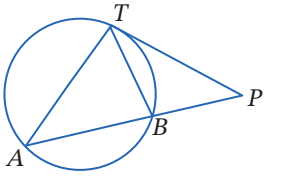
2 2 circles with centres P and Q touch at T with a common tangent AB. R C



a Explain why $\angle ATP$ and $\angle ATQ$ are right angles.

b Hence, show that P, T and Q are on the same line.

3 PT is a tangent and PA is a secant. R C



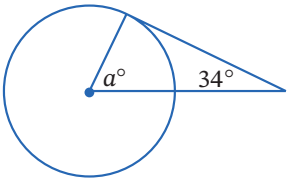
a Prove that $\triangle PTA \,|||\, \triangle PBT$.

b Hence, show that $\frac{PT}{PB} = \frac{PA}{PT}$ and $PT^2 = PB \times PA$.

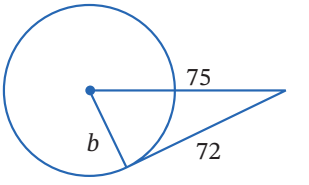
EXAMPLE 7

4 Find the value of each variable, giving reasons. R C

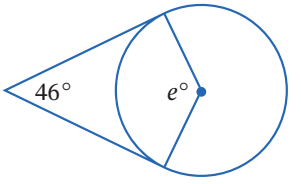
a



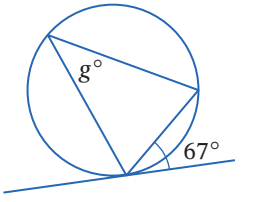
b



c



d



Foundation Standard Complex

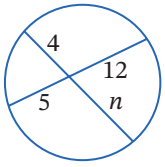
9780170479660

5 Find the value of n. R C

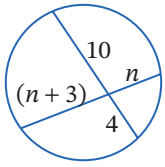
PATH

EXAMPLE 8

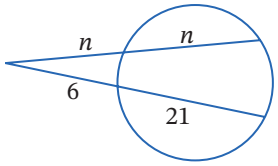
a



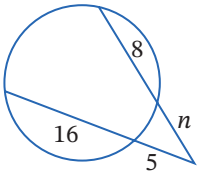
b



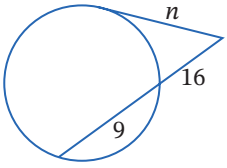
c



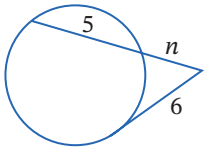
d



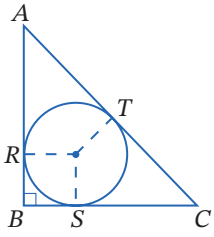
e



f

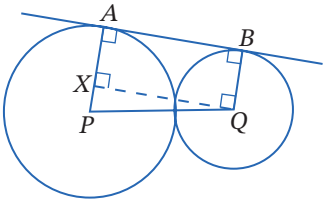


6 ABC is a right-angled triangle with an internal circle such that each side is a tangent to the circle. R, S and T are the respective points of contact of each of the tangents. If $SC = 30$ cm, $RB = 5$ cm and $AT = x$, find the value of x. R C



7 2 circles with centres P and Q and radii 18 cm and 8 cm respectively, touch each other externally as shown. Find the length of: R C

a XP **b** AB



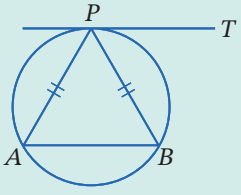
Proofs using circle theorems 15.05

Example 9

PATH

PT is a tangent to a circle, and $PA = PB$.

Prove that $PT \parallel AB$.



Solution

$\angle TPB = \angle A$ (angle in alternate segment)

$\angle A = \angle B$ (equal angles of isosceles $\triangle PAB$)

$\therefore \angle TPB = \angle B$

$\therefore PT \parallel AB$ (alternate angles are equal)

Foundation Standard Complex

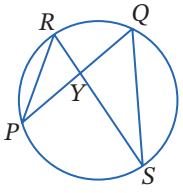
PATH

EXERCISE 15.05 ANSWERS ON P. 689

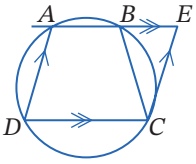
Proofs using circle theorems U F R C

EXAMPLE 9

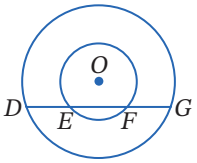
1 PQ and RS are 2 chords of a circle intersecting at the point Y. Prove that the triangles PYR and SYQ are similar and hence $PY \times YQ = RY \times YS$. R C



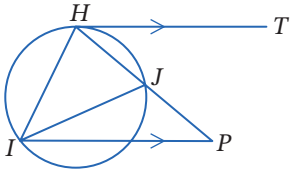
2 $ABCD$ is a cyclic quadrilateral. Side AB is produced to E so that $AECD$ is a parallelogram. Prove that $\triangle CBE$ is an isosceles triangle. R C



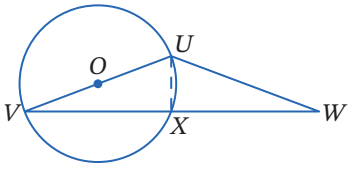
3 O is the centre of 2 concentric circles. $DEFG$ is a straight line. Prove that $DE = FG$. R C



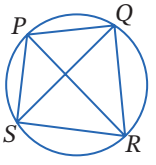
4 HT is a tangent and parallel to IP. HJP is a straight line. Prove that $\angle HIP = \angle HJI$. R C



5 UVW is an isosceles triangle, where $UV = UW$. A circle with centre O is drawn on one of the equal sides. Prove that the circle bisects the base of the triangle. R C

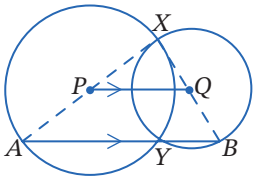


6 $PQRS$ is a cyclic quadrilateral. PR bisects $\angle QRS$. Prove that $\triangle SPQ$ is isosceles. R C

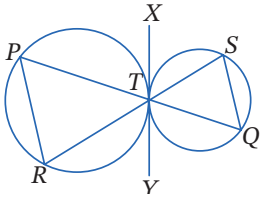


7 2 circles with centres P and Q intersect at X and Y. AB is parallel to PQ, the interval joining the centres of the circles.

Prove that $PQ = \frac{1}{2}AB$. R C



8 2 circles touch externally at T. XY is the common tangent. PTQ and RTS are straight lines. Prove that $PR \parallel SQ$. R C



Foundation Standard Complex

9780170479660